



VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

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Slot : L23+L24

Course : Exploratory Data Analysis Lab

Experiment 7

Code

```
credit <- read.csv("C:\\Users\\vaibh\\Downloads\\Credit Card Customer Data.csv")
```

```
cat("STRUCTURE OF DATASET\\n")
```

```
str(credit)
```

```
cat("\\nSUMMARY OF DATASET\\n")
```

```
summary(credit)
```

```
cat("\\nMISSING VALUES IN EACH ATTRIBUTE\\n")
```

```
print(colSums(is.na(credit)))
```

```
credit_num <- credit[, sapply(credit, is.numeric)]
```

```
if("CUST_ID" %in% names(credit_num))
```

```
{
```

```

credit_num <- subset(credit_num, select = -CUST_ID)
}

for(i in 1:ncol(credit_num))
{
  credit_num[is.na(credit_num[, i]), i] <- mean(credit_num[, i], na.rm = TRUE)
}

credit_scaled <- scale(credit_num)

k <- as.integer(readline(prompt = "Enter the value of k: "))

set.seed(123)

kmeans_result <- kmeans(credit_scaled,
  centers = k,
  nstart = 25)

cat("\nCLUSTER CENTERS\n")
print(kmeans_result$centers)

cat("\nNUMBER OF RECORDS IN EACH CLUSTER\n")
print(kmeans_result$size)

cat("\nWITHIN CLUSTER SUM OF SQUARES (WCSS)\n")
print(kmeans_result$tot.withinss)

```

```

pca_result <- prcomp(credit_scaled)

pca_data <- data.frame(
  PC1 = pca_result$x[,1],
  PC2 = pca_result$x[,2],
  Cluster = as.factor(kmeans_result$cluster)
)

plot(pca_data$PC1,
     pca_data$PC2,
     col = pca_data$Cluster,
     pch = 19,
     xlab = "Principal Component 1",
     ylab = "Principal Component 2",
     main = "K-Means Clusters Visualized Using PCA")

legend("topright",
     legend = levels(pca_data$Cluster),
     col = 1:length(levels(pca_data$Cluster)),
     pch = 19,
     title = "Cluster")

```

Output

